WORKSHEET

Chapter 8: Compound interest investments with regular deposits

A financial solver is required to complete this worksheet.

1 At the start of each year, Carwin deposits $3000 into an account earning 9% p.a. interest that is compounded annually. Interest is paid at the end of the year.

a Write a recurrence relation that gives the balance of the account at the end of year n, B_n.

b What will be the value of the investment at the end of the 10th year?

2 At the start of 2011, Pip opened an investment account with a $1000 deposit. The account earns 8.4% p.a. interest compounded annually. Interest is paid at the end of each year. Pip makes a $700 deposit into the account at the start of every following year.

a Write a recurrence relation that gives the balance of the account at the end of year n, B_n.

b What will be the value of the investment at the end of 2017?

3 Natasha opens an account for her retirement savings that earns 11.4% p.a. interest, compounded annually. She initially deposits $20 000. Every year thereafter, she invests a further $15 000 into the account.

a Write a recurrence relation that gives the balance of the account at the end of year n, B_n.

b How much interest will be earned in the first year?

c How long will it be before Natasha's savings exceed $300 000?

4 Exactly 2 years before he finishes school, Horace opens a new savings account with a $500 deposit. The account earns 6.5% p.a. interest, compounded monthly. He is using the account to save for a holiday. He works regular shifts at the local supermarket and invests $300 at the end of each month into the account.

a Write a recurrence relation that gives the balance of the account at the end of year n, B_n.

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b How much money will Horace have in his holiday account when he finishes school?

5 Marvolo wants to save $3000 in 1 year by making equal monthly deposits into a bank account that earns 12.4% p.a. interest, compounded monthly.

a How much does Marvolo need to deposit into his account each month in order to achieve his goal?

b How much interest will his investment have earned over the year?

6 Minerva is saving to go on a $50 000 luxury cruise in 4 years. She opens a savings account with $10 000 and wants to make equal monthly deposits at the start of every following month to reach her goal. If the interest rate on the account is 9.5% p.a. compounded monthly, what should be her monthly deposit?

7 Reine has a savings account that earns 5.9% p.a. interest, compounded monthly. With the balance at $5500, she wants to start a new savings regimen in which she deposits the same amount into the account at the start of every month for a year. Interest is added to the account at the end of the month. Her aim is to have saved a total of $8000 by the end of the year. If her first deposit is to be made immediately:

a how much does she need to deposit each month to achieve this goal

b what will be the balance of the account at the end of the third month?

8 Cameron has saved $20 000. He is saving for a deposit on a house and would like to save a total of $100 000. His savings account provides monthly compound interest at 7.6% p.a. At the end of each month for the next 3 years he makes equal deposits into the account. How much does he need to deposit into the account each month to achieve his goal?

9 Mandy starts a European holiday savings account. She deposits $400 into it at the start of each month.

a How long will it take Mandy to save $25 000 if interest is compounded monthly on the account at 6.9% p.a.?

b How much total interest will Mandy's investment earn?

10 Orlando has decided he will retire when his savings reach $700 000. His current savings come to a total of $550 000 and are in an account earning 8.3% p.a. interest, compounded monthly. If he deposits $3500 into the account at the end of each month, immediately after the interest is added, how long will it be before he can retire?

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Answers

1 **a** $B_{n+1} = 1.09(B_n + 3000)$, where $B_1 = 3270$

b $49 680.88

2 **a** $B_{n+1} = 1.084(B_n + 700)$, where $B_1 = 1084$

b $7381.70

3 **a** $B_{n+1} = 1.114(B_n + 15\,000)$, where $B_1 = 22\,280$

b $2280

c 11 years

4 **a** $B_{n+1} = 1.00542(B_n + 300)$, where $B_1 = 502.71$

b $7936.39

5 **a** $236.11

b $166.68

6 $626.69

7 **a** $175.72

b $6113.89

8 $1858.84

9 **a** 54 months or 4 and a half years

b $3400

10 20 months or 1 year and 8 months

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